EN3160: Image Processing and Machine Vision – Assignment 2

Fitting and Alignment

Department of Electronic and Telecommunication Engineering, University of Moratuwa | October 2026

Name: Samarasinghe S.M.R.R. | **Index:** 230566U **GitHub:** RumethRanhinda/EN3160-Vision-UoM

1. Blob Detection using Laplacian of Gaussians

1.1 Justification of Scale (σ) and Radius Parameters

The image is loaded with `cv.IMREAD_REDUCED_COLOR_4`, reducing its dimensions to 1/4th of its size (360×360 pixels). In this resolution:

- The smallest discernible sunflower centers in the background near the horizon span a radius of $r_{\min} \approx 2$–3 px. Since the scale-normalized LoG response peaks at $r = \sqrt{2}\sigma$, the lower scale bound is $\sigma_{\min} = r_{\min}/\sqrt{2} \approx 2/\sqrt{2} \approx 1.50$. Meanwhile, The largest foreground sunflowers span a radius of $r_{\max} \approx 10$–11.3 px, yielding the upper scale bound $\sigma_{\max} = r_{\max}/\sqrt{2} \approx 11.31/\sqrt{2} \approx 8.00$.
- A geometric scale progression $\sigma_i = \sigma_{\min} k^i$ with $S = 20$ levels ($k = 1.092$) was chosen, spanning radii $r \in [2.12, 11.31]$ px to provide uniform logarithmic coverage across scales.

1.2 Methodology

- **Scale-Normalized LoG:** For each scale σ_i, the response $\sigma_i^2 \nabla^2 (G_{\sigma_i} * I)$ is computed. Sunflower centers are dark circular disks against bright yellow petals, giving positive extremas.
- **3D Extrema Detection:** Local maxima are detected in a $3 \times 3 \times 3$ spatio-scale volume. A point is marked as a blob if it is strictly greater than all 26 neighbors and exceeds contrast threshold $\tau = 0.10$. Scale boundaries are excluded.

1.3 Important Code

```python
import cv2, numpy as np, scipy.ndimage as ndi

# 1. Scale-space generation using scale-normalized LoG
sigmas = np.geomspace(1.5, 8.0, 20)  # Justified sigma range
scale_space = np.zeros((len(sigmas), *gray.shape))
for i, s in enumerate(sigmas):
    scale_space[i] = (s ** 2) * ndi.gaussian_laplace(gray, sigma=s)

# 2. 3D Non-Maximum Suppression across (scale, y, x)
max_vol = ndi.maximum_filter(scale_space, footprint=np.ones((3, 3, 3)), mode='constant')
peaks = (scale_space == max_vol) & (scale_space > 0.10)
peaks[[0, -1], :, :] = False  # Exclude scale boundary slices

# 3. Extract circle parameters: center (x, y) and radius r = sqrt(2) * sigma
blobs = []
for s_idx, y, x in np.argwhere(peaks):
    if 0 < x < gray.shape[1] - 1 and 0 < y < gray.shape[0] - 1:
        s = sigmas[s_idx]
        blobs.append({'center': (float(x), float(y)), 'r': np.sqrt(2)*s,
                      'sigma': s, 'score': scale_space[s_idx, y, x]})
```

1.4 Results



Figure 1: Detected circles (yellow) with top 5 largest circles highlighted (red).

Rank	Center (x, y)	Radius r (px)	Scale σ	LoG Resp.
1	$(152.0, 254.0)$	10.36	7.33	0.1328
2	$(282.0, 338.0)$	10.36	7.33	0.3492
3	$(254.0, 174.0)$	9.49	6.71	0.2398
4	$(310.0, 174.0)$	9.49	6.71	0.1760
5	$(79.0, 258.0)$	9.49	6.71	0.1777

Table 1: Parameters of the 5 largest detected circles.

Summary of Results:

- **Range of σ Values Used:** $\sigma \in [1.50, 8.00]$ across 20 geometrically spaced levels ($k = 1.092$), covering radii from 2.12 px to 11.31 px.
- **Parameters of Largest Circles:** Reported in Table 1. Restricting $\sigma_{\max} = 8.00$ isolates individual flower heads (e.g., $r = 10.36$ px at $(282.0, 338.0)$) without over-merging into background trees.
- **Total Detections:** A total of 314 blobs were detected across the image, successfully marking sunflower centers from the distant field to the foreground.